

WEEK 1

Task1

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Source Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main(){
    char command[100];
    pid_t pid;
    printf("Enter a Linux command: ");
    scanf("%s", command);
    pid = fork();
    if(pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if(pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());

        execlp(command, command, NULL);
        printf("Command execution failed.\n");
        exit(1);
    } else {
        wait(NULL);
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child process completed.\n");
    }
    return 0;
}
```

Output:

```
Enter a Linux command: ls

Child Process
Child PID : 437
OSSP code.c files.c gm.c hello.c os program.c programming.c source_code
code files gm hello operating os.c programming programming.c.save source_code.c

Parent Process
Parent PID : 436
Child process completed.
```

Task2

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Report

1. Uname: Displays operating system and kernel information.

```
shivam@MSI123:~$ uname
Linux
shivam@MSI123:~$ |
```

2. lscpu: Displays CPU information.

```

shivam@MSI123:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       12
  On-line CPU(s) list:       0-11
Vendor ID:                    GenuineIntel
Model name:                   12th Gen Intel(R) Core(TM) i5-12450H
CPU family:                   6
Model:                        154
Thread(s) per core:           2
Core(s) per socket:           6
Socket(s):                    1
Stepping:                     3
BogoMIPS:                     4991.99
Flags:                         fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                             r pge mca cmov pat pse36 clflush mmx fxsr sse s
                             se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                             _tsc rep_good nopl xtopology tsc_reliable nonst
                             op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                             sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                             opcnt tsc_deadline_timer aes xsave avx f16c rdr
                             and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                             brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                             d ept_ad fsgsbase tsc_adjust bml avx2 smep bmi
                             2 erms invpcid rdseed adx smap clflushopt clwb
                             sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
                             vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
                             vdiri movdir64b fsrm md_clear serialize ibt flu
                             sh_lld arch_capabilities

Virtualization features:
  Virtualization:             VT-x
  Hypervisor vendor:           Microsoft
  Virtualization type:         full
Caches (sum of all):
  L1d:                         288 KiB (6 instances)
  L1i:                         192 KiB (6 instances)
  L2:                          7.5 MiB (6 instances)

```

```

sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnmi umip waitpkg gfni vaes vpclmulqdq rdpid mo
vdiri movdir64b fsrmd md_clear serialize ibt flu
sh_lld arch_capabilities

Virtualization features:
  Virtualization: VT-x
  Hypervisor vendor: Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d: 288 KiB (6 instances)
  L1i: 192 KiB (6 instances)
  L2: 7.5 MiB (6 instances)
  L3: 12 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Vulnerable: No microcode
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
shivam@MSI123:~$

```

3. lsblk: Displays storage devices.

```

shivam@MSI123:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda         8:0    0 356.9M  1 disk
sdb         8:16    0 159.4M  1 disk
sdc         8:32    0    2G    0 disk [SWAP]
sdd         8:48    0    1T    0 disk /snap
                                     /mnt/wslg/distro
                                     /
shivam@MSI123:~$

```

4. ps: Displays currently running processes.

```
shivam@MSI123:~$ ps
  PID TTY          TIME CMD
  382 pts/0    00:00:00 bash
  626 pts/0    00:00:00 ps
shivam@MSI123:~$
```

5. top: Displays real-time system performance.

```
shivam@MSI123:~$ top
top - 09:40:06 up 4 min, 1 user, load average: 0.06, 0.08, 0.03
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.
MiB Mem : 7794.6 total, 6468.8 free, 542.7 used, 934.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7252.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+
1	root	20	0	21628	13156	9796	S	0.0	0.2	0:00.41
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.01
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00
52	root	19	-1	42164	16232	15156	S	0.0	0.2	0:00.09
99	root	20	0	25336	6736	5124	S	0.0	0.1	0:00.12
111	root	20	0	152944	1768	1512	S	0.0	0.0	0:00.00
114	root	20	0	153076	1656	1396	S	0.0	0.0	0:00.00
116	root	20	0	153076	1828	1512	S	0.0	0.0	0:00.00
125	root	20	0	153076	1704	1444	S	0.0	0.0	0:00.00
130	root	20	0	153076	1680	1424	S	0.0	0.0	0:00.00
135	root	20	0	526764	12536	1536	S	0.0	0.2	0:01.90
160	systemd+	20	0	21460	13192	10932	S	0.0	0.2	0:00.06
162	systemd+	20	0	91028	7944	6976	S	0.0	0.1	0:00.04
217	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.00
218	message+	20	0	9632	5304	4600	S	0.0	0.1	0:00.03
227	root	20	0	2147016	42272	24680	S	0.0	0.5	0:00.37
231	root	20	0	17964	8852	7812	S	0.0	0.1	0:00.04
233	root	20	0	1756104	12996	10888	S	0.0	0.2	0:00.07
247	syslog	20	0	222516	5976	4572	S	0.0	0.1	0:00.04
256	root	20	0	3124	1976	1836	S	0.0	0.0	0:00.00
270	root	20	0	107020	23188	13728	S	0.0	0.3	0:00.08
380	root	20	0	3196	1120	980	S	0.0	0.0	0:00.00
381	root	20	0	3196	1252	1108	S	0.0	0.0	0:00.01
382	shivam	20	0	7568	6144	4372	S	0.0	0.1	0:00.03
383	root	20	0	6704	4648	3864	S	0.0	0.1	0:00.00
427	shivam	20	0	20376	11652	9492	S	0.0	0.1	0:00.06
428	shivam	20	0	21152	3576	1852	S	0.0	0.0	0:00.00
454	shivam	20	0	6080	5360	3692	S	0.0	0.1	0:00.01
629	shivam	20	0	9300	5760	3572	R	0.0	0.1	0:00.01

